Department of Mathematics and Computer Science

Legal Lean

Innovation project by Simon Holm, Johannes Rothe & Andreas Bonde

Introduction (1 min)

- din mor

Introduction



FORM but for Law

- MathLib and CSLib as gotten a lot of succes
- Provide a structured database for Law
- Provide a way for AI be trained better

Introduction

LEGAL
LEAN

FORM but for Law
- MathLib and CSLib as gotten a lot of succes
- Provide a structured database for Law
- Provide a way for AI be trained better

Header

- din mor

Department of Mathematics and Computer Science

Lean

Challenge

Select a pre-defined challenge

```
theorem eulers_four_square_identity (a b c d p q r s : Int) :
    (a^2 + b^2 + c^2 + d^2) * (p^2 + q^2 + r^2 + s^2) =
      (a*p - b*q - c*r - d*s)^2 + (a*q + b*p + c*s - d*r)^2
      + (a*r - b*s + c*p + d*q)^2
      + (a*s + b*r - c*q + d*p)^2 := by
  sorry
```

Candidate Solution

```
import Std

/-- Induction and SMT-like reasoning using `grind` -/
theorem geometric_seris_sum (x : Int) (n : Nat) :
    (x - 1) * ((List.range n).map (x ^ ·)).sum
      = x ^ n - 1 := by
  induction n <;> grind [List.range_succ]

/-- algebraic reasoning in using `grind` -/
theorem eulers_four_square_identity (a b c d p q r s : Int) :
    (a^2 + b^2 + c^2 + d^2) * (p^2 + q^2 + r^2 + s^2) =
      (a*p - b*q - c*r - d*s)^2 + (a*q + b*p + c*s - d*r)^2
      + (a*r - b*s + c*p + d*q)^2
      + (a*s + b*r - c*q + d*p)^2 := by
  grind

/-- `bv_decides` decides bitvector equations -/
theorem power_of_two_bit_test (x : BitVec 64) :
    (x != 0 && x &&& (x - 1) == 0) = (x.cpop == 1) := by
  bv_decide
```

Success. Comparator verified that the solution proves the claims described in the challenge's theorem (`eulers_four_square_identity`), and the challenge is recognized.

sdu.dk

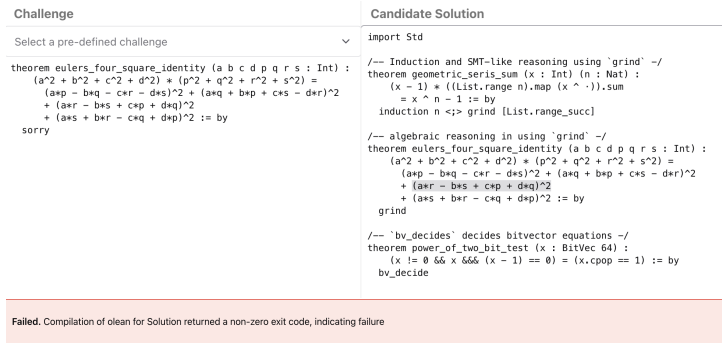
#sdudk



Header

- din mor

Lean

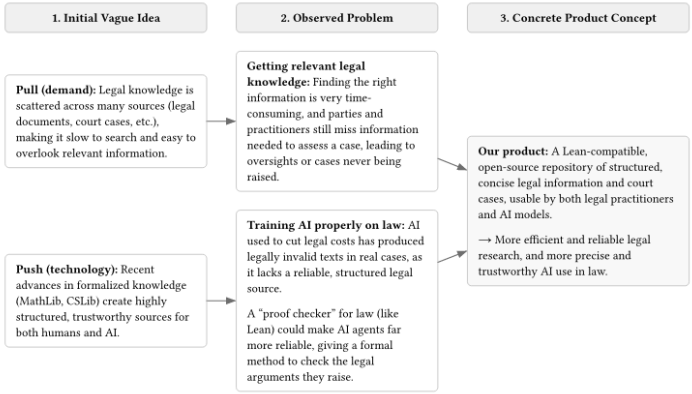




Header

- din mor

Idea Funnel


1. Initial Vague Idea
2. Observed Problem
3. Concrete Product Concept
Pull (demand): Legal knowledge is scattered across many sources (legal documents, court cases, etc.), making it slow to search and easy to overlook relevant information.
Getting relevant legal knowledge: Finding the right information is very time-consuming, and parties and practitioners still miss information needed to assess a case, leading to oversights or cases never being raised.
Our product: A Lean-compatible, open-source repository of structured, concise legal information and court cases, usable by both legal practitioners and AI models.
→ More efficient and reliable legal research, and more precise and trustworthy AI use in law.
Push (technology): Recent advances in formalized knowledge (MathLib, CSLib) create highly structured, trustworthy sources for both humans and AI.
Training AI properly on law: AI used to cut legal costs has produced legally invalid texts in real cases, as it lacks a reliable, structured legal source.
A "proof checker" for law (like Lean) could make AI agents far more reliable, giving a formal method to check the legal arguments they raise.


Idea Funnel

Stakeholder Mapping

Direct Users	Organisational Stakeholders	External / Enabling Stakeholders
• Legal practicioners • Legal practicioners using AI • Compliance Staff • AI engineers	• Academic staff as maintainers • Student maintainers • Other contributors • Core maintainer team (review and merge rights) • Legal expert reviewers who approve formalizations • CI / infrastructure maintainers	• Government • Big companies • Universities • Civilstyrelsen / Retsinformation (publisher of Danish law) • Domstolsstyrelsen / Domsdatabasen (publisher of verdicts) • Datatilsynet (GDPR authority) • Advokatsamfundet (bar association) • The Lean / Mathlib community and FORM at SDU • Law faculties at universities • Github

Stakeholder Mapping



SWOT Analysis

Direct Users	Organisational Stakeholders	External / Enabling Stakeholders
• Legal practicioners • Legal practicioners using AI • Compliance Staff • AI engineers	• Academic staff as maintainers • Student maintainers • Other contributors • Core maintainer team (review and merge rights) • Legal expert reviewers who approve formalizations • CI / infrastructure maintainers	• Government • Big companies • Universities • Civilstyrelsen / Retsinformation (publisher of Danish law) • Domstolsstyrelsen / Domsdatabasen (publisher of verdicts) • Datatilsynet (GDPR authority) • Advokatsamfundet (bar association) • The Lean / Mathlib community and FORM at SDU • Law faculties at universities • Github

SWOT Analysis
Direct Users
• Legal practicioners
• Legal practicioners using AI
• Compliance Staff
• AI engineers
Organisational Stakeholders
• Academic staff as maintainers
• Student maintainers
• Other contributors
• Core maintainer team (review and merge rights)
• Legal expert reviewers who approve formalizations
• CI / infrastructure maintainers
External / Enabling Stakeholders
• Government
• Big companies
• Universities
• Civilstyrelsen / Retsinformation (publisher of Danish law)
• Domstolsstyrelsen / Domsdatabasen (publisher of verdicts)
• Datatilsynet (GDPR authority)
• Advokatsamfundet (bar association)
• The Lean / Mathlib community and FORM at SDU
• Law faculties at universities
• Github

Talk with a professor

Direct Users	Organisational Stakeholders	External / Enabling Stakeholders
• Legal practicioners • Legal practicioners using AI • Compliance Staff • AI engineers	• Academic staff as maintainers • Student maintainers • Other contributors • Core maintainer team (review and merge rights) • Legal expert reviewers who approve formalizations • CI / infrastructure maintainers	• Government • Big companies • Universities • Civilstyrelsen / Retsinformation (publisher of Danish law) • Domstolsstyrelsen / Domsdatabasen (publisher of verdicts) • Datatilsynet (GDPR authority) • Advokatsamfundet (bar association) • The Lean / Mathlib community and FORM at SDU • Law faculties at universities • Github

Talk with a professor

